

Network Cables and connectors

Task (1):



Task (2):

Network Cable	Real-World Application / Device	Why It Is Used
Cat5e (Category 5e)	Wi-Fi Router	Cat5e is affordable and supports speeds up to 1 Gbps , making it suitable for connecting a home Wi-Fi router to a modem or network device.
Cat6 (Category 6)	Zomato POS Terminal	Cat6 provides faster speeds and better resistance to interference than Cat5e, giving POS terminals a reliable wired connection for processing orders and payments.
Fiber Optic Cable	IPL Streaming Server	Fiber optic cables transmit data using light at very high speeds and over long distances. They are ideal for handling the large amount of data required for live IPL video streaming .

Task (3):

Steps to Crimp an RJ-45 Connector

1. Cut the cable – Cut the Cat5e/Cat6 cable to the required length using a cable cutter.
2. Remove the outer jacket – Use a wire stripper to carefully remove about 2–3 cm of the outer insulation without damaging the internal wires.
3. Untwist the wires – Separate the four twisted pairs and straighten the individual wires.
4. Arrange the wires – Follow the T568B color sequence:
5. White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
6. Trim the wires – Hold the wires together in the correct order and cut their ends evenly.
7. Insert into the RJ-45 connector – Push the wires into the connector until all eight wires reach the end and remain in the correct pin order.
8. Crimp the connector – Place the RJ-45 plug into a crimping tool and squeeze firmly to lock the contacts onto the wires.
9. Test the cable – Connect both ends to a cable tester and check that all eight pins have the correct connection.

Two Common Mistakes to Avoid

- Incorrect wire color order: Mixing up the T568B sequence can make the cable fail or cause incorrect pin connections.
- Wires not inserted fully: If the wires do not reach the end of the RJ-45 plug, the metal contacts may not make proper contact.

Task (4):

Cable and Connector Selection for an IPL Screening Gaming Zone

For an IPL screening event with multiple screens and gaming consoles, I would use HDMI 2.1 cables with HDMI connectors for connecting consoles to screens, and Cat6 Ethernet cables with RJ-45 connectors for networking the consoles and other devices.

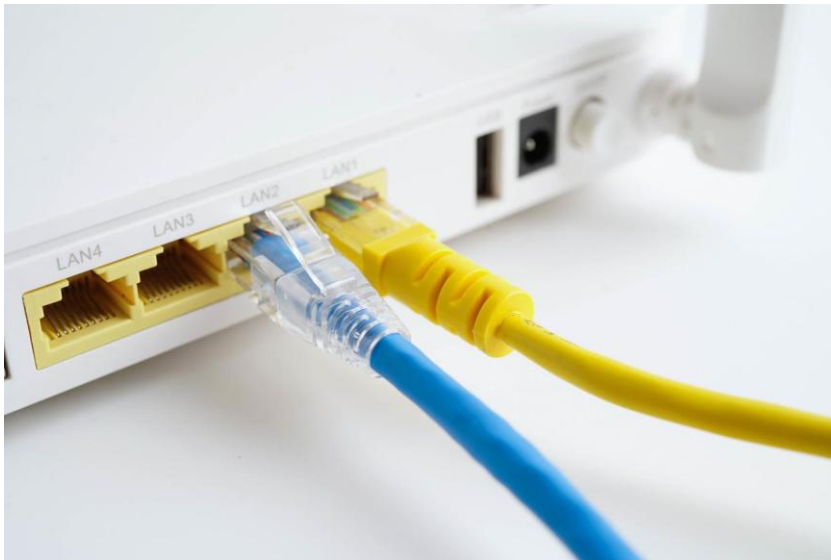
Cable	Connector	Use	Reason
HDMI 2.1	HDMI	Connect gaming consoles to TVs/monitors	Supports high-resolution video such as 4K and high refresh rates, providing smooth gaming and high-quality IPL video.
Cat6 Ethernet	RJ-45	Connect consoles, PCs, switches and routers	Provides a fast, stable wired network connection with lower interference and latency than Wi-Fi.
Fiber Optic	LC/SC	Connect the main network to distant areas	Useful for long-distance, high-bandwidth connections when the gaming zone has many devices or is spread across a large venue.

Recommended Setup:

Gaming Console → HDMI 2.1 → Screen/TV

Console/PC → Cat6 + RJ-45 → Network Switch → Router/Internet

Task (5):



- **Cable type:** Ethernet cable (Cat5e/Cat6)
- **Connector:** RJ-45 connector
- **Devices connected:** A Wi-Fi router and a computer/network device or modem.
- **Purpose:** The Ethernet cable provides a wired network connection for transferring data and accessing the Internet. Wired Ethernet is generally more stable and less affected by wireless interference than Wi-Fi.